

Samuel Lavictoire

(613) 360-3982 • samuellavictoire1@gmail.com • [Project Portfolio](#)

ENGINEERING EXPERIENCE

Chassis & Suspension Lead | Mars Rover University Engineering Team 2024 – Present

- Led the suspension design for a 50-member team, ranking 3rd best Canadian team at CIRC (2024)
- Designed and assembled suspension systems including machined and composites parts, ensuring manufacturability and structural integrity
- Ran ANSYS FEA simulations to verify structural integrity under competition load cases, reducing component mass by 6–8 kg compared to the prior design iteration
- Created detailed engineering drawings for machined parts and 3D-printed components
- Coordinated a sub-team of 15 engineers across concurrent suspension, drivetrain, and wheel research sub-tasks

CubeSat ADCS and Propulsion systems | Senior Capstone Project 2025

- Designed attitude determination and control system (ADCS) incorporating custom-machined reaction wheels and a commercial magnetic torquer, selected through a trade study to meet mission pointing requirements
- Sized and designed a cold gas propulsion system including a custom propellant tank and nozzle to meet the required thrust of the mission
- Integrated both subsystems within the allocated CubeSat volume in collaboration with structures, payload, and power sub teams

Nozzle CFD Analysis | Final project 2026

- Implemented an axisymmetric solver for the 2D compressible Euler equations in C++ using the Local Lax Friedrichs flux function
- Validated the Cold gas nozzle design and compared the numerical values results to the isentropic nozzle equations.

PROFESSIONAL EXPERIENCE

Mechanical/Buildings Engineering Intern | CIMA+ May – Aug 2024

- Reconciled multi-revision MEP drawing sets for federal and commercial building retrofits, cross-referencing fit-up and demolition changes by date across HVAC, plumbing, and electrical disciplines to produce a single current-state plan while following PSPC CAD standards
- Compiled site inspection records, contractor notes, and photographic documentation into structured progress reports submitted to clients for financing and investor reporting purposes

Energy Efficiency & Technology – Co-op Student | Natural Resources Canada Sept – Dec 2023

- Reviewed Energy Star certification applications for commercial buildings, evaluating submitted HVAC system designs and energy performance data against program criteria
- Researched HVAC system configurations and distilled findings into concise technical summaries for non-engineering stakeholders

Energy Grid Modelling Co-op Student | Environment and Climate Change Canada Jan – Apr 2023

- Automated code version tracking and output formatting using Python and Pandas, improving team workflow efficiency on an active energy grid modelling project
- Contributed to the team's documentation with automated Git-based version control and research on energy technologies

EDUCATION

BASc Mechanical Engineering & BSc Computing Technology | University of Ottawa 2020 – May 2026

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, AutoCAD, Ansys FEA, CFD

Manufacturing: Manual machining (lathe & mill), 3D printing (FDM), composites layup & epoxy bonding

Programming: Python, C/C++, Java